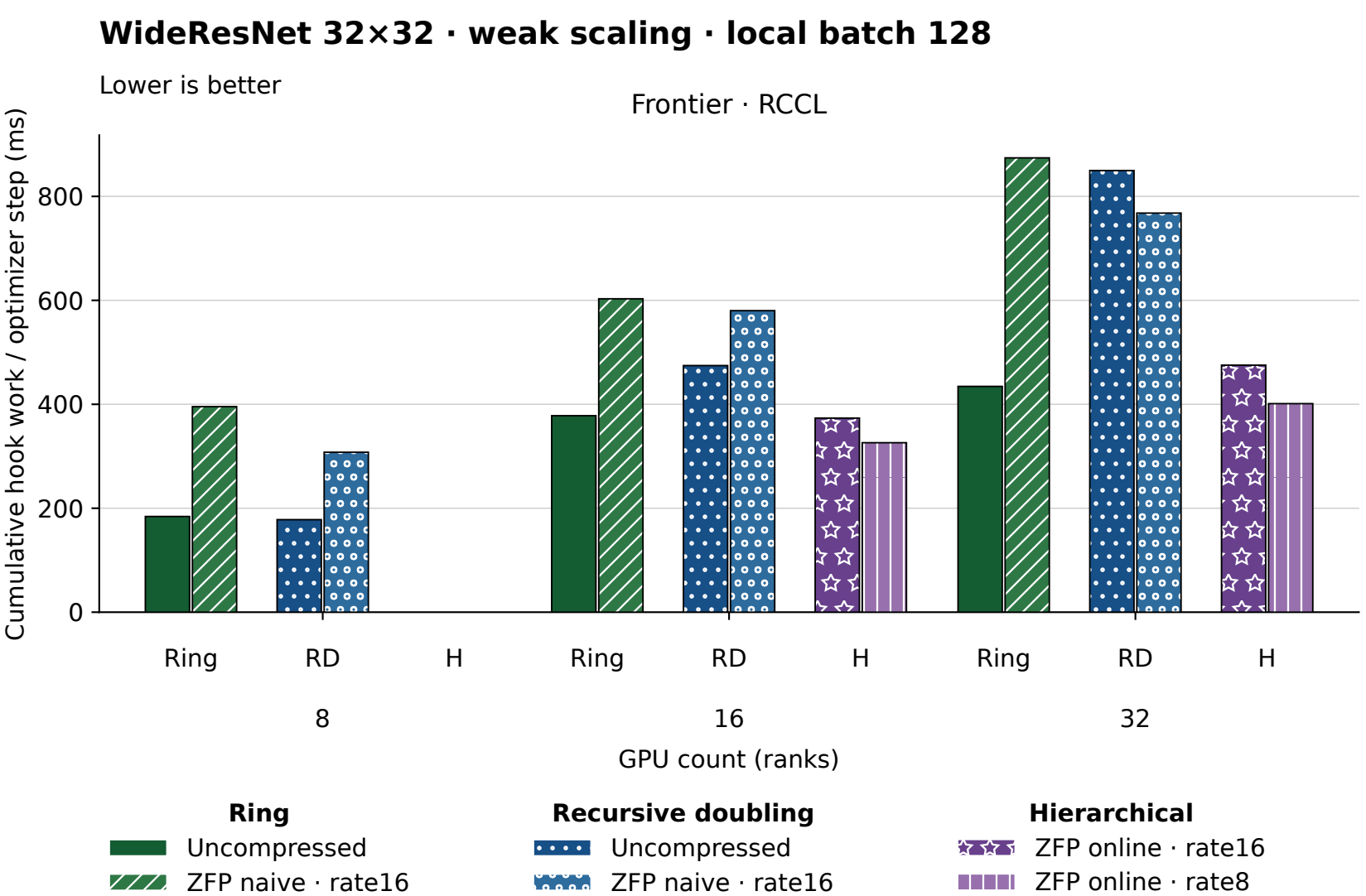

WideResNet 32×32 · weak scaling · local batch 128
Lower is better
Frontier · RCCL
Cumulative hook work / optimizer step (ms)
0
200
400
600
800
Ring
RD
H
Ring
RD
H
Ring
RD
H
8
16
32
GPU count (ranks)
Ring
Uncompressed
ZFP naive · rate16
Recursive doubling
Uncompressed
ZFP naive · rate16
Hierarchical
ZFP online · rate16
ZFP online · rate8
Epoch 1 excluded · missing metrics are not zero · r = ZFP rate